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CSCI 4962/6962: Frontiers in Blockchain Research

Bitcoin (continued) and Timestamping Documents

Class 03
Jan 20, 2026

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Announcement

- My office hour this week (Friday 12pm-1pm) in Lally 306 is canceled.
- Course mentor Caleb's office hours this week is Wed 4 pm – 5 pm ET.
    - Tentative location: Library.
    - The exact location will be announced by Caleb on Submitty.

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Discussion points from the Bitcoin whitepaper

- What problem is Bitcoin *actually* solving?
  - Is Bitcoin primarily about:
    - Electronic cash?
    - Eliminating trusted intermediaries?
    - Solving distributed consensus under adversarial conditions?
- What *wasn't* solved by earlier digital cash systems?
- Would Bitcoin still matter if banks were perfectly trustworthy?

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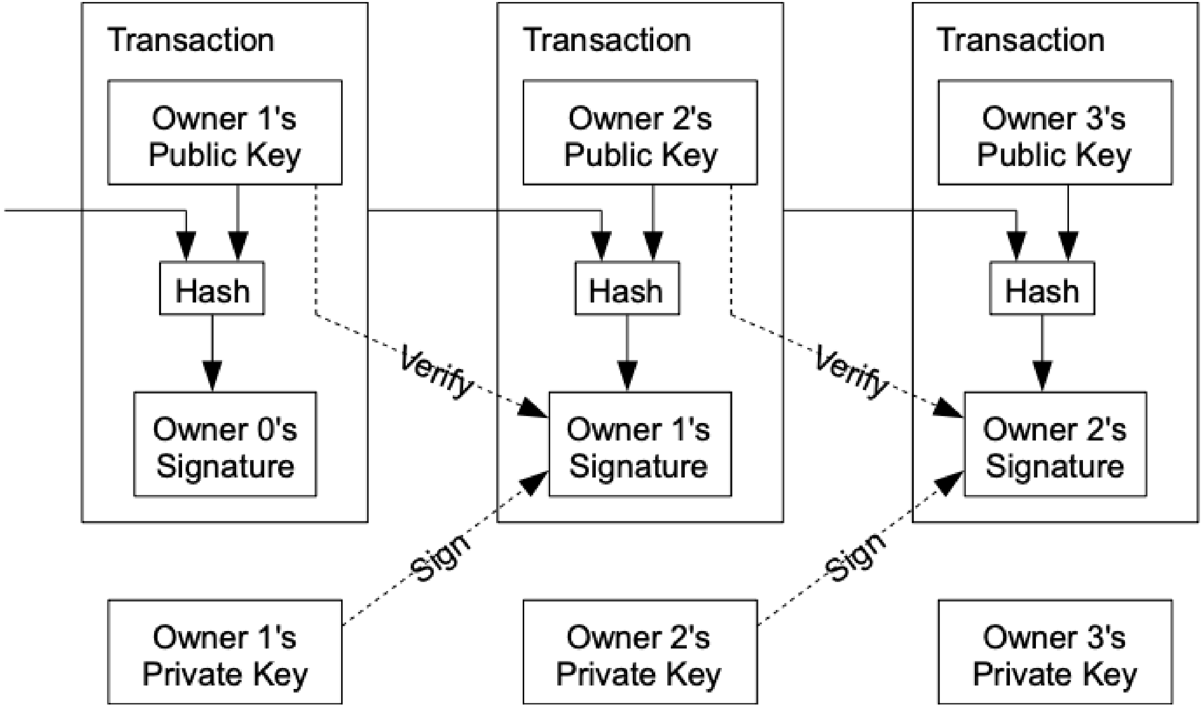
Electronic coin as a chain of digital signatures



The challenge

- Digital signatures verify ownership, but they do not prove a coin hasn’t been spent already.
- Common solution:
  - A central Mint checks every transaction (centralized)
- Bitcoin solution:
  - Transactions must be publicly announced.
  - Participants agree on a single history of the order in which they were received.

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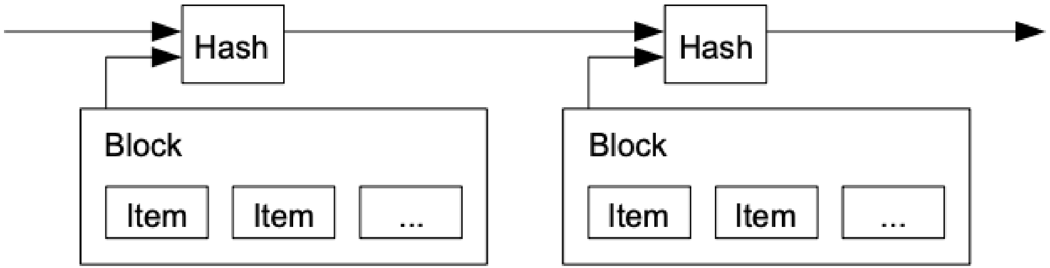
Establishing Time: The Timestamp Server



- A timestamp server takes a hash of a block of items and publishes it.
- Each timestamp includes the previous timestamp in its hash, forming a chain, with each additional timestamp reinforcing the ones before it.

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Consensus Mechanisms

- Consensus is an agreement among a group of people on an idea, statement, or plan of action
  - Majority: 51%
  - Supermajority: 66% (sometimes higher)
  - Unanimous: 100%
  - Weighted: not all votes weighed equally or multiple votes per agent
- Consensus is typically only relevant when there is no distinguished leader
  - A jury must reach a consensus on a court verdict (unanimous)
  - The senate must reach a consensus on new bills being passed (majority or supermajority)
- Particularly important when there is significant disagreement or potential for untrustworthy parties in the discussions around the decision

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What is Consensus in Blockchain?

- The **agreement** of system components (nodes) on the **[next] state** of the system, or the **transition** between the current state and the next state
  - The nodes must agree on a set of **valid transactions** representing the change from the current state of the system to the next state of the system
  - Consensus must be **achieved automatically** (without human oversight)
- Consensus is **irreversible**: posted transactions are final
- Blockchain consensus is a subset of distributed system consensus
  - Distributed System: A number of independent computers linked by a network
- Must be resistant to malicious or false actors
- For permissioned blockchain networks, legal remedies may be used if a user acts maliciously.

The Need for Consensus

- Consensus is a very difficult problem when parties are not trusted
- The network must maintain integrity in order to maintain value
- Past transactions must be trusted for the network to function
- Thus, the ability to verify transactions without trust is needed
- The consensus problem can often be rephrased as **the ability to trust the result of a transaction or block, without trusting the parties involved in the transaction, or the party that verified it**

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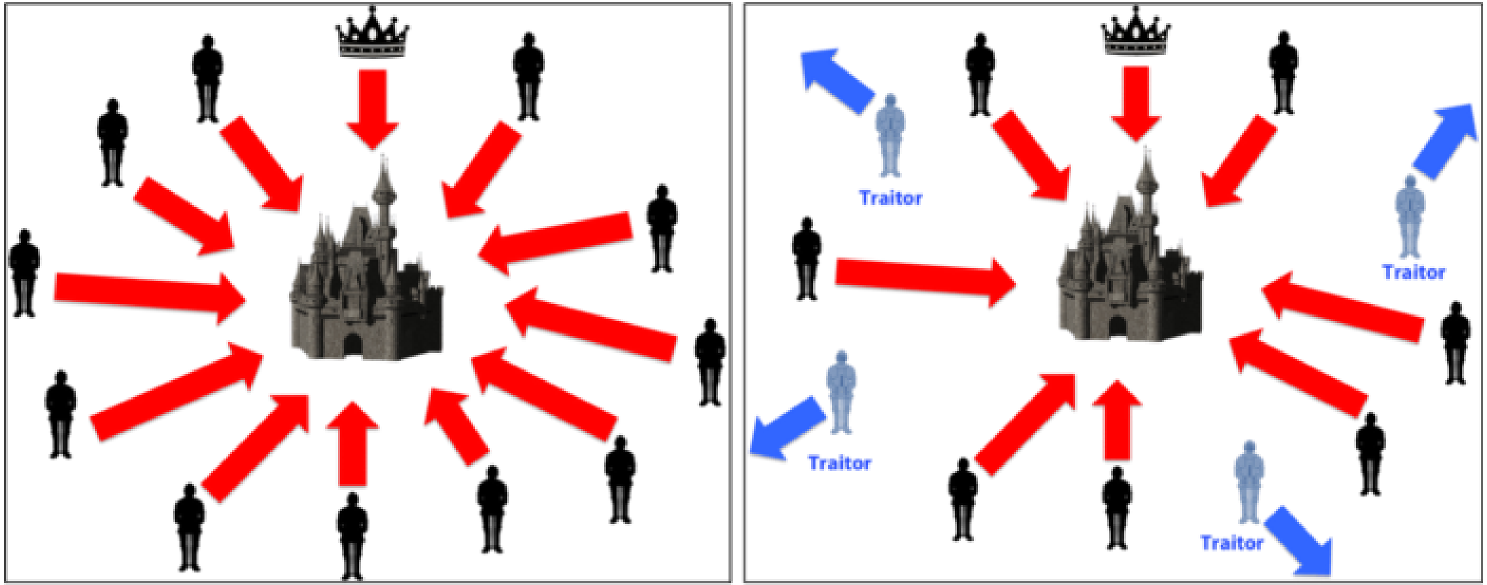
The Byzantine Generals Problem



Coordinated Attack Leading to Victory **Uncoordinated Attack Leading to Defeat**

Source: https://iconpinas.medium.com/blockchain-basics-consensus-algorithm-part-3-dfd339644f10

Byzantine Fault Tolerance (BFT)

- A system is considered to be Byzantine Fault Tolerant (BFT) if it is **resistant to the dilemmas of the Byzantine Generals Problem**
- A Byzantine Fault is defined as a failure mode of the system, either **failing to function** or **functioning incorrectly**, caused by an **inability to reach or error in consensus among the system**
- Most consensus algorithms used in blockchain technologies are Byzantine Fault Tolerant

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Proof-of-Work (PoW) / Nakamoto Consensus

PoW is computational proof that some work has been performed.

Usually, this is implemented through a computational puzzle.
The puzzle should be:

- easy to state *(find me a needle in this haystack)*
- hard to solve *(the haystack is big)*
- easy to verify *(yep, that's a needle)*

Generating a solution to the puzzle serves as evidence that the solver performed some work.

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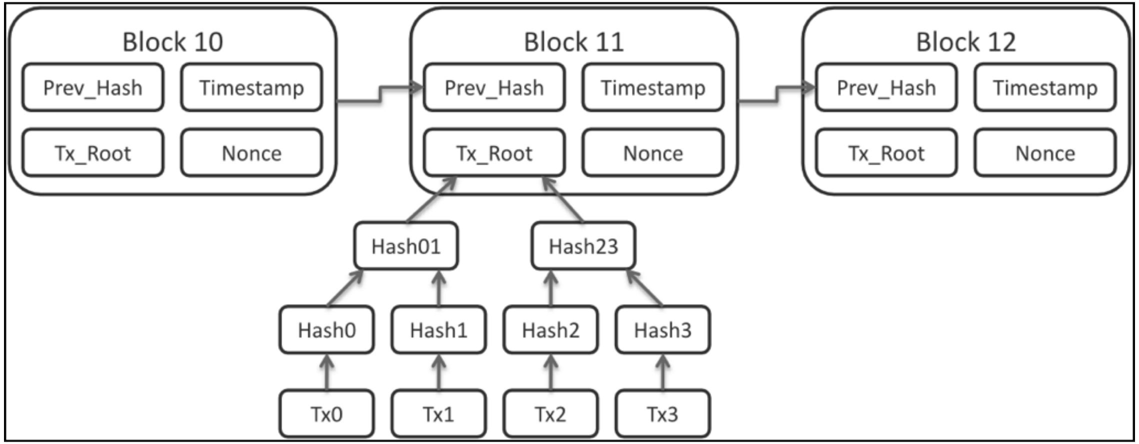
Proof of Work (PoW) / Nakamoto Consensus

- A given node collects transactions that are broadcast to the entire network and stores them in a block
  - Before including transactions in the block, the node verifies that the transactions are valid
  - Invalid transactions result in a block being rejected by the other nodes

- The transactions are typically assembled in a type of Merkle tree



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Proof of Work (PoW) / Nakamoto Consensus

- The transactions pay a fee to the mining node to be included the transaction, higher fees are included first
- The mining node begins solving an extremely difficult cryptographic hashing problem, with the transactions being part of the input to the problem
  - This is essentially a guessing game with a very low chance of guessing correctly
  - Once the correct answer is known, it is very easy for other people to check that it works
  - Part of the motivation for solving the problem is the rewards the miner receive

Example “Computational Puzzle”

- The computer must find a hash value meeting the following target criteria (known as the difficulty level):

```
SHA256(“blockchain” + Nonce) = Hash Digest starting with “000000”
```

- The text string “blockchain” is appended with a nonce value, and then the hash digest is calculated.
- The nonce values used will be numeric values only.

```
SHA256("blockchain0") =
0xbd4824d8ee63fc82392a6441444166d22ed84eaa6dab11d4923075975acab938 (not solved)

SHA256("blockchain1") =
0xdb0b9c1cb5e9c680dfff7482f1a8efad0e786f41b6b89a758fb26d9e223e0a10 (not solved)

...

SHA256("blockchain10730895") =
0x000000ca1415e0bec568f6f605fcc83d18cac7a4e6c219a957c10c6879d67587 (solved)
```

- Each additional “leading zero” value increases the difficulty.

Proof of Work (PoW) / Nakamoto Consensus

- The mining node that has found the correct solution broadcasts it to the rest of the network, and begins the next block with another complex cryptographic hashing problem
- The longest blockchain (weighted by work) is always taken to be the correct chain, and thus the other miners will also begin the new problem: the length of each block is determined by how much work it took to create
- The other nodes can quickly check that the transactions included in the block are valid, and that the broadcast solution is actually a solution to the problem
  - Verification of the nonce is easy since only a single hash needs to be done to check to see if it solves the puzzle.

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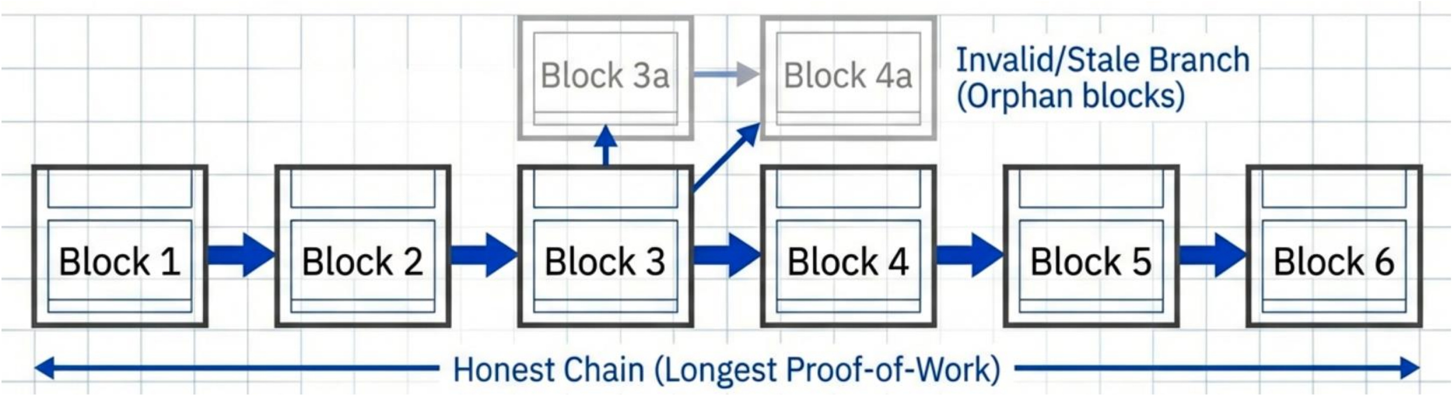
The Longest Chain Rule: One CPU One Vote



The voting problem: if the majority were based on one IP address one vote, it could be subverted by anyone allocating many IPs. (Sybil Attack)
The solution: PoW is essentially one CPU one vote.
Consensus: The majority decision is represented by the longest chain, which has the greatest PoW invested in it.

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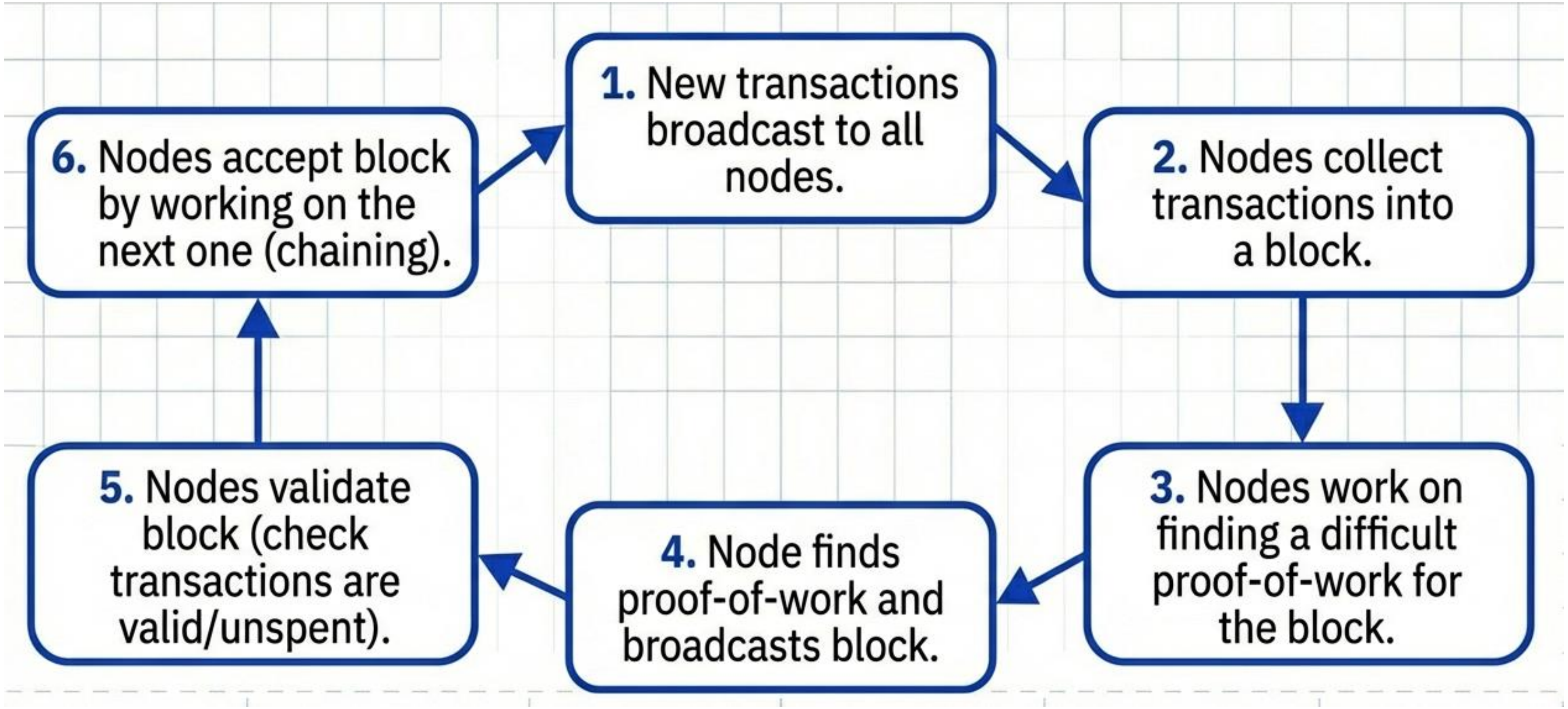
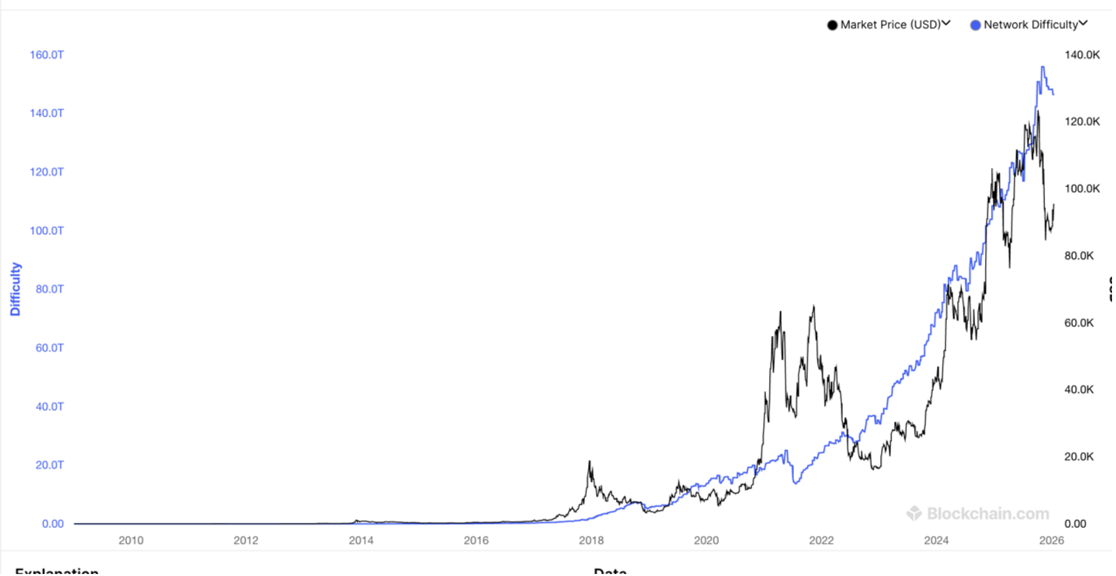
Network Operations



Bitcoin Network Difficulty



The difficulty is a measure of how to find a hash below a given target. A high difficulty means that it will take more computing power to mine the same number of blocks, making the network more secure against attacks.

The difficulty is adjusted every 2016 blocks (every 2 weeks approximately) so that the average time between each block remains 10 minutes.

Source: https://www.blockchain.com/charts/difficulty

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Bitcoin Mining Hashrate

Source: https://www.blockchain.com/explorer/charts/hash-rate

- Mining hashrate is a key security metric.
- The more hashing (computing) power in the network, the greater its security and its overall resistance to attack.
- Although Bitcoin's exact hashing power is unknown, it is possible to estimate it from the number of blocks being mined and the current block difficulty.

Bitcoin Halving

- A Bitcoin halving event is when the reward for mining Bitcoin transactions is cut in half.
- When Bitcoin first started, 50 Bitcoins per block were given as a reward to miners.
- After every 210,000 blocks are mined (approximately every 4 years), the block reward halves and will keep on halving until the block reward per block becomes 0 (approximately by year 2140).
- This event also cuts in half Bitcoin's inflation rate and the rate at which new Bitcoins enter circulation.
- Previous halvings have correlated with intense boom and bust cycles that have ended with higher prices than prior to the event.
- Bitcoin last halved on 05 May 2024 12:27:18 UTC, resulting in a block reward of 3.125 BTC.

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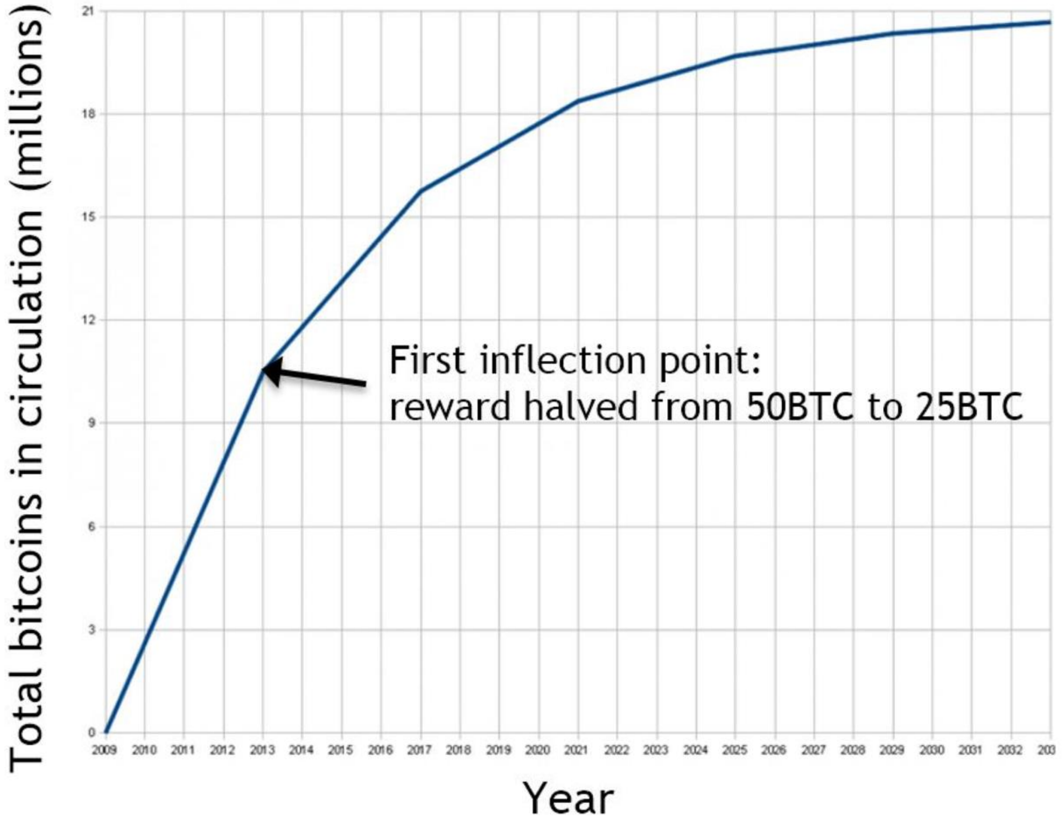
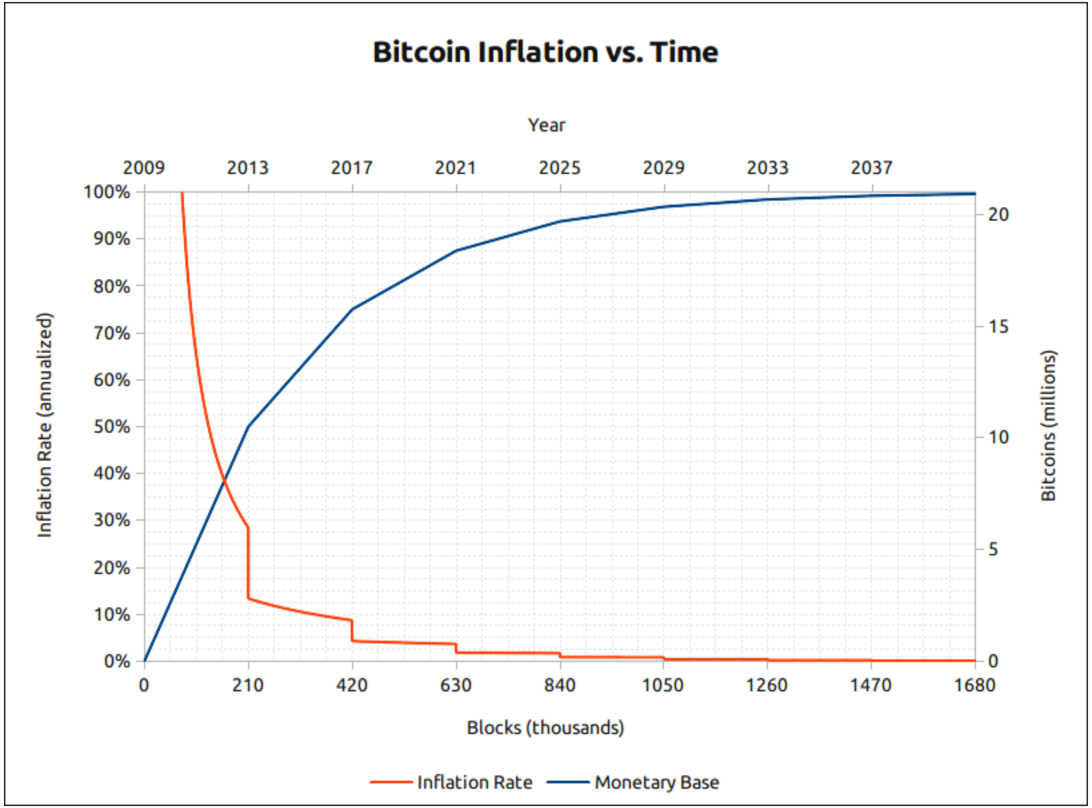
PoW Incentives with Bitcoin as an Example



Bitcoin Halving



Source: https://www.bitcoinblockhalf.com/

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Mining Pools

- Nodes organize themselves into “pools” or “collectives,” whereby they work together to solve puzzles and split the reward.
- Work can be distributed between two or more nodes across a collective to share the workload and rewards
- For example:
  - `Node 1: check nonce 0000000000 to 0536870911`
  - `Node 2: check nonce 0536870912 to 1073741823`
  - `Node 3: check nonce 1073741824 to 1610612735`
  - `Node 4: check nonce 1610612736 to 2147483647`
- What about “Sybil Attacks”?
  - A Sybil attack is a computer security attack (not limited to blockchain networks) where an attacker can create many nodes (i.e., creating multiple identities) to gain influence and exert control.
  - PoW combats this by having the focus of **network influence** being the **amount of computational power** (hardware, which costs money) mixed with a **lottery system** (the most hardware increases likelihood but does not guarantee it).

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PoW Strengths

- Proven reliability
- Predictable block times
- Robust, and does not rely on any other node being trustworthy
- Cannot be censored
    - Public transactions can be seen as a drawback in some cases

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PoW Drawbacks

- The need for specialized hardware, i.e., Application Specific Integrated Circuit (ASIC)
- Energy Costs
- The only known vulnerability is the so-called ‘**51% attack**’
  - One miner or group of miners is able to take over the resources driving the chain forward.
  - However, expenditure of large computing and energy cost to take 51% would be lost if crypto collapsed. Historical example, **alchemists**.
- Mining pools are able to slightly game the system by **delaying the release of a found block** and gaining a slight head start on the next one
- Scalability issues
  - Lower transaction throughput
  - Lowering the block time (problem difficulty) is potentially less secure
- Miners often sell the coins immediately, removing any loyalty to the chain they are mining

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Proof-of-Work as a *security–economics* mechanism

- Why is **computational cost** essential to consensus?
- How does PoW convert *energy* into *security*?
- What attacks become expensive, but not impossible?

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How to time-stamp a digital document

Stuart Haber and W. Scott Stornetta

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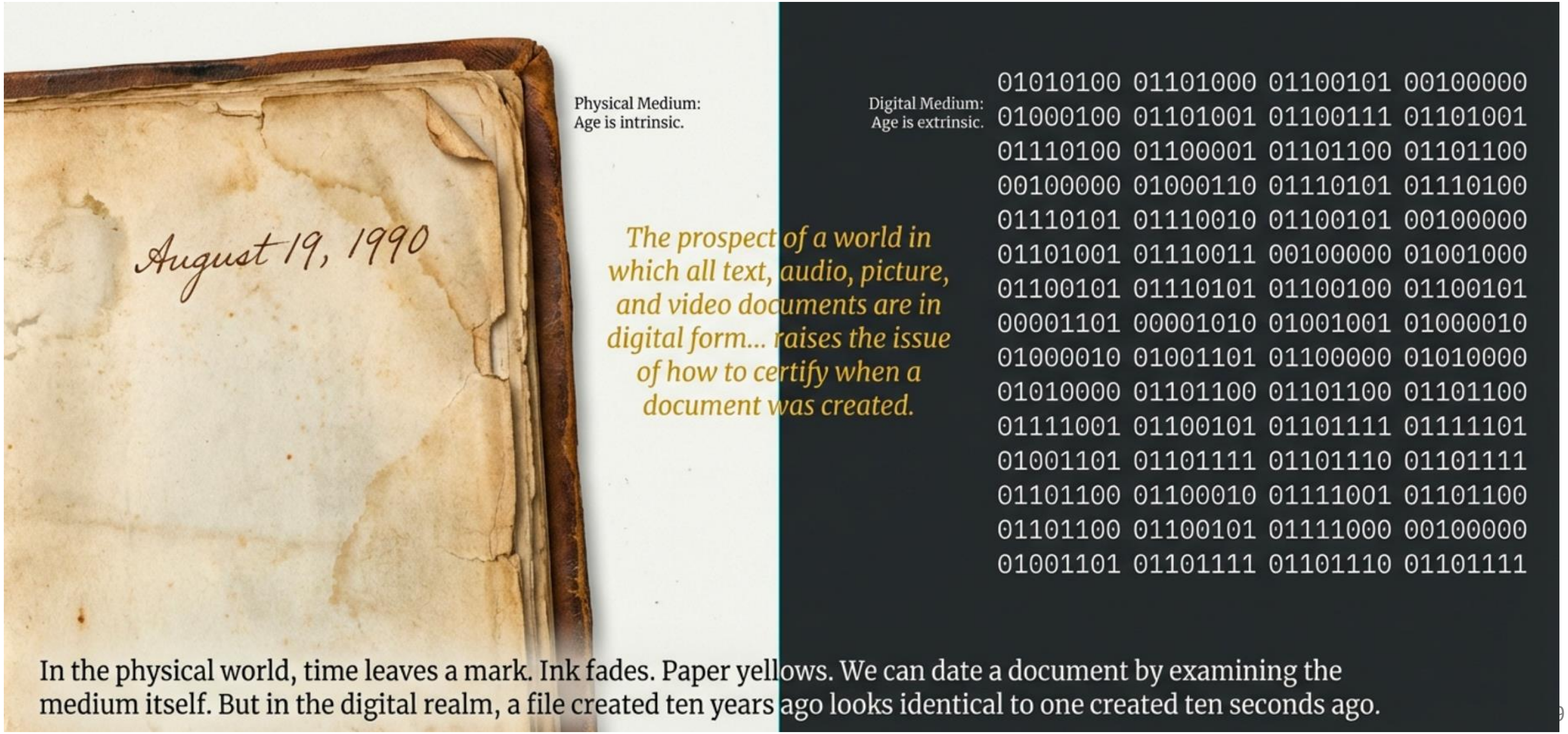
Erasure of History



The Problem: Data vs. Medium

Traditional dating methods rely on the container (carbon dating paper, analyzing magnetic decay).

But digital files are fluid.

They move between servers and drives without changing their internal structure.

We must shift our focus from the container to the content.

We must solve the problem of time-stamping the data, not the medium.

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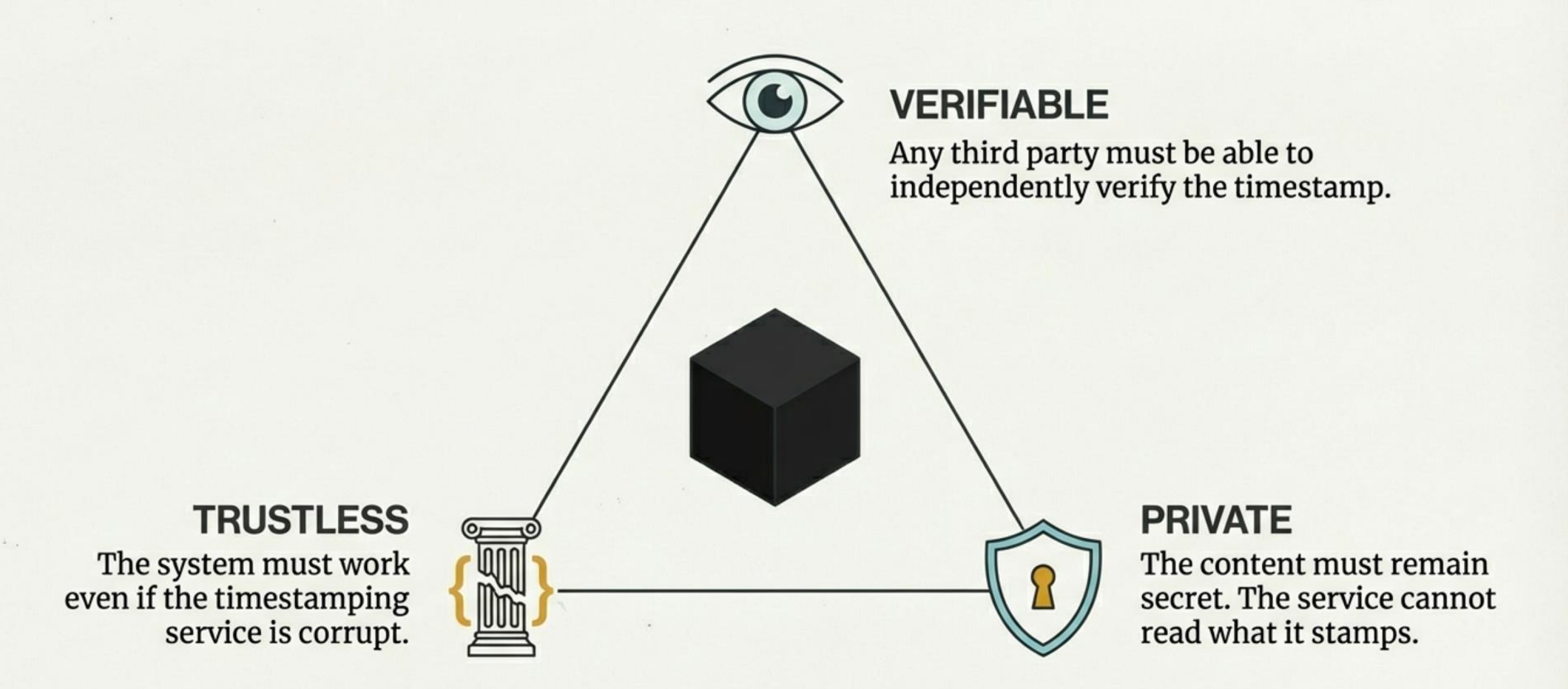
The Constraints of Certification



A robust solution must satisfy all three simultaneously.

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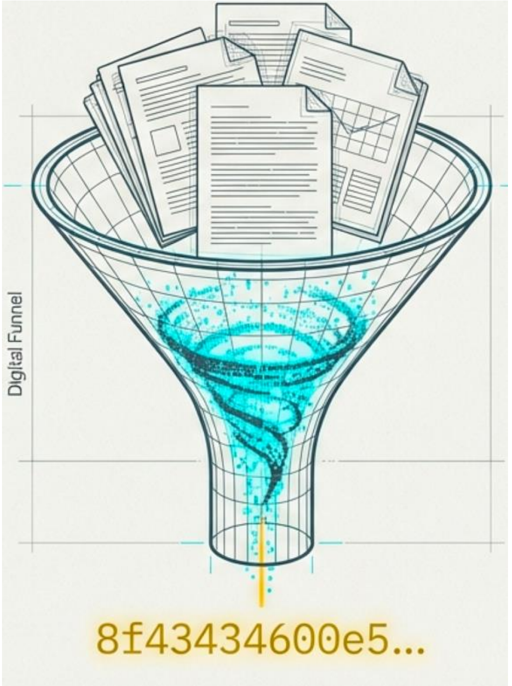
The Tool: Digital Fingerprints

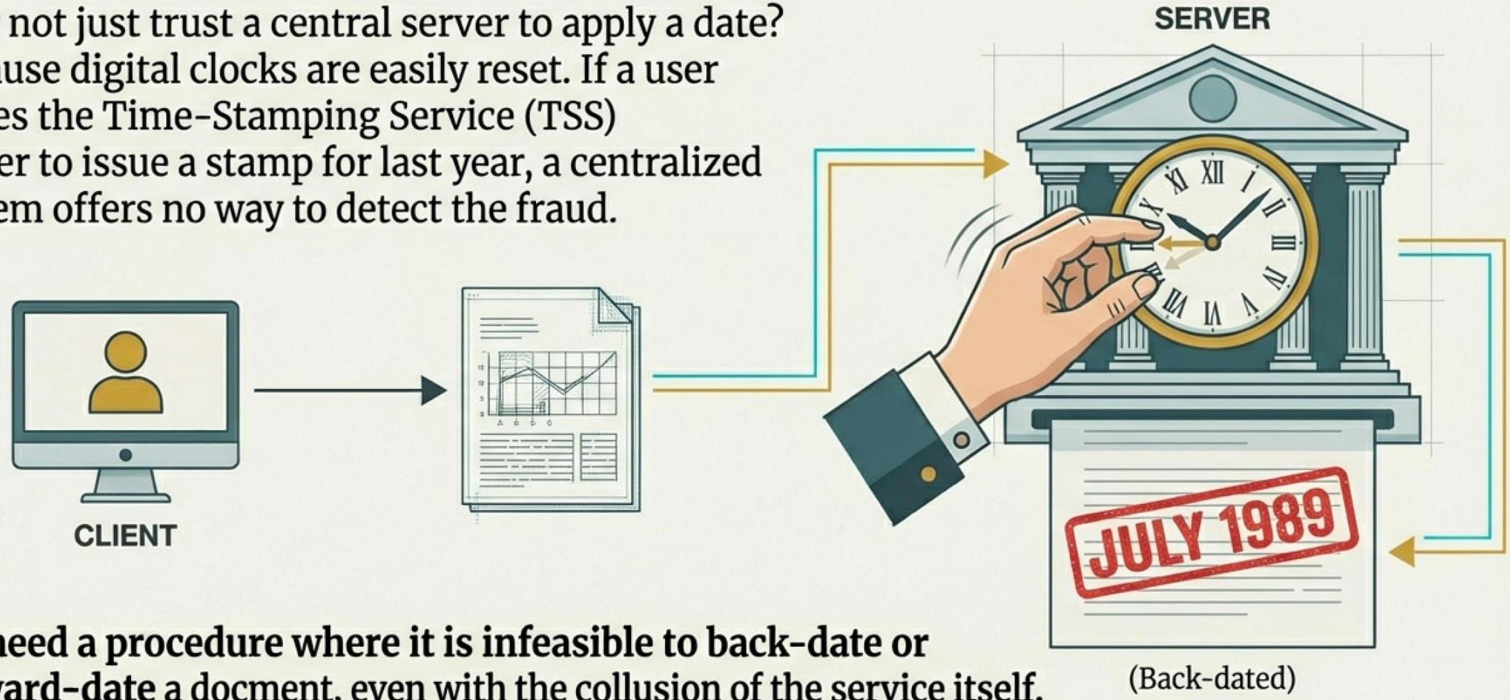
- One-way hash function to create a unique identifier for every document.
- This compresses the data and ensures the specific version of the document is fixed in time.

- **The Rule:**
  - Timestamp the hash, not the document.



The Threat: The Colluding Server

Why not just trust a central server to apply a date? Because digital clocks are easily reset. If a user bribes the Time-Stamping Service (TSS) owner to issue a stamp for last year, a centralized system offers no way to detect the fraud.



We need a procedure where it is infeasible to back-date or forward-date a docment, even with the collusion of the service itself.

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The Solution: Linked Integrity

- To prevent collusion, we must abandon the isolated events. Instead, we establish a sequence.
- Every document “fixes” the history of the one before it.
- There is a chain of evidence.

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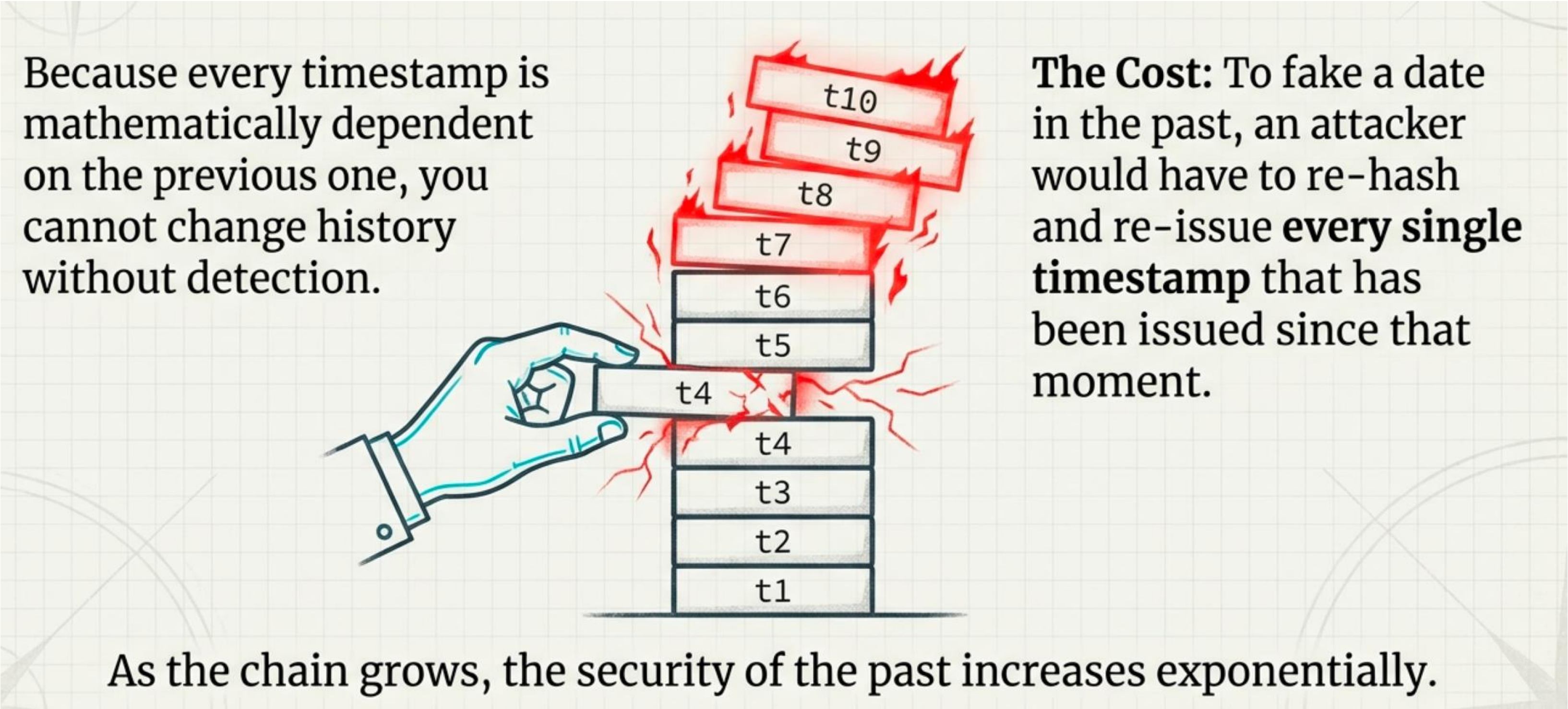
Computational Infeasibility



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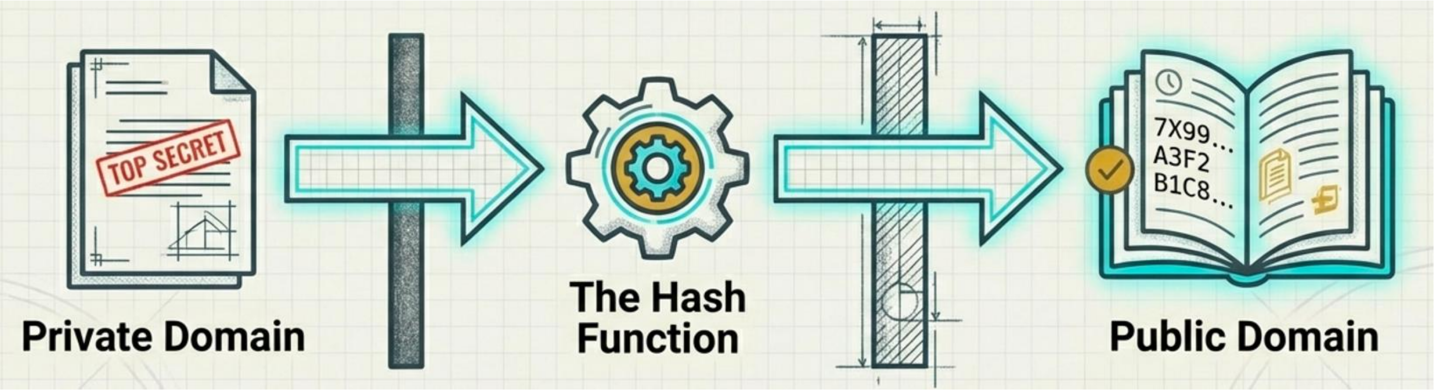
Distributed Trust and Zero-Knowledge Verification

- The validity of the chain doesn’t rely on the word of the TSS.
- It relies on the widely distributed record of hashes held by various users.
- You can publicly prove that you owned an idea at a specific time without revealing what the idea was.



The Haber-Stornetta Protocol

1. **HASH GENERATION:** The client uses a one-way hash function to calculate the digital fingerprint of the document.
2. **TRANSMISSION:** The client sends *only* the hash to the Time-Stamping Service (TSS).
3. **CERTIFICATION:** The TSS appends the current date/time and the hash of the *immediately preceding* request.
4. **SIGNATURE:** The TSS digitally signs the new composite bundle (Certificate).
5. **VERIFICATION:** If challenged, any party can traverse the chain of links to verify the sequence is unbroken.

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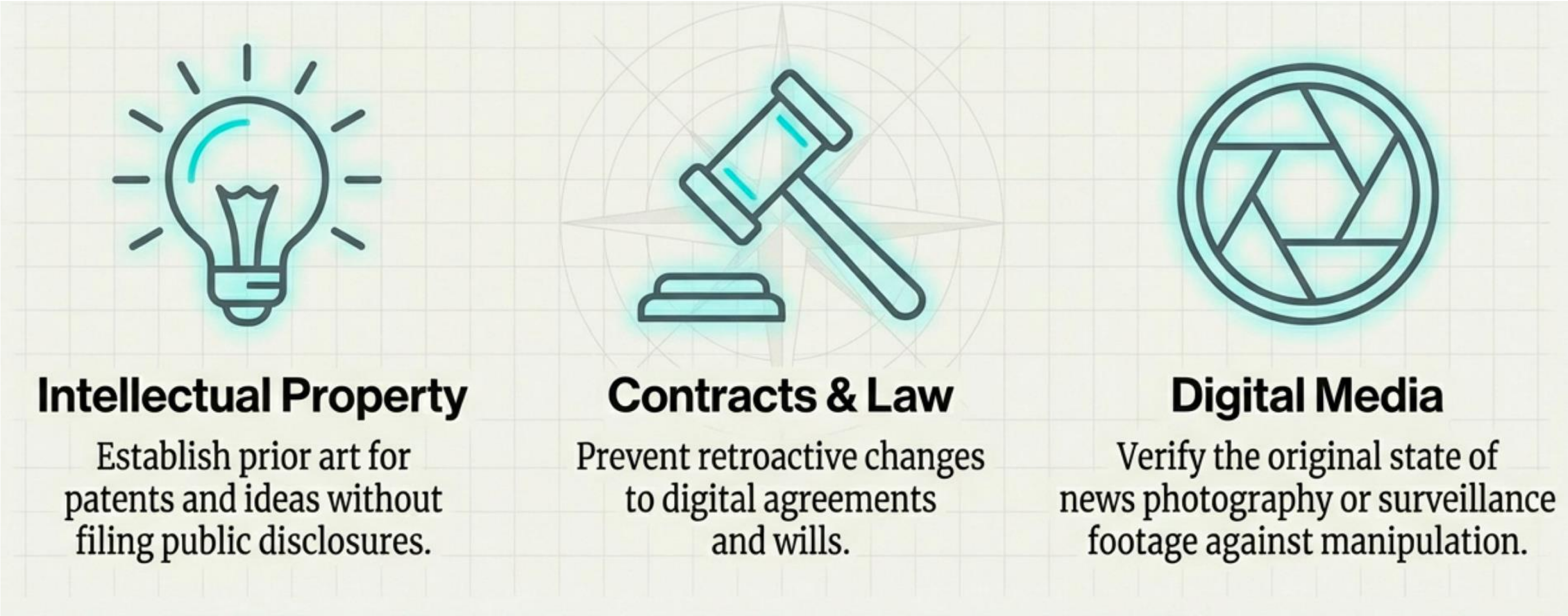
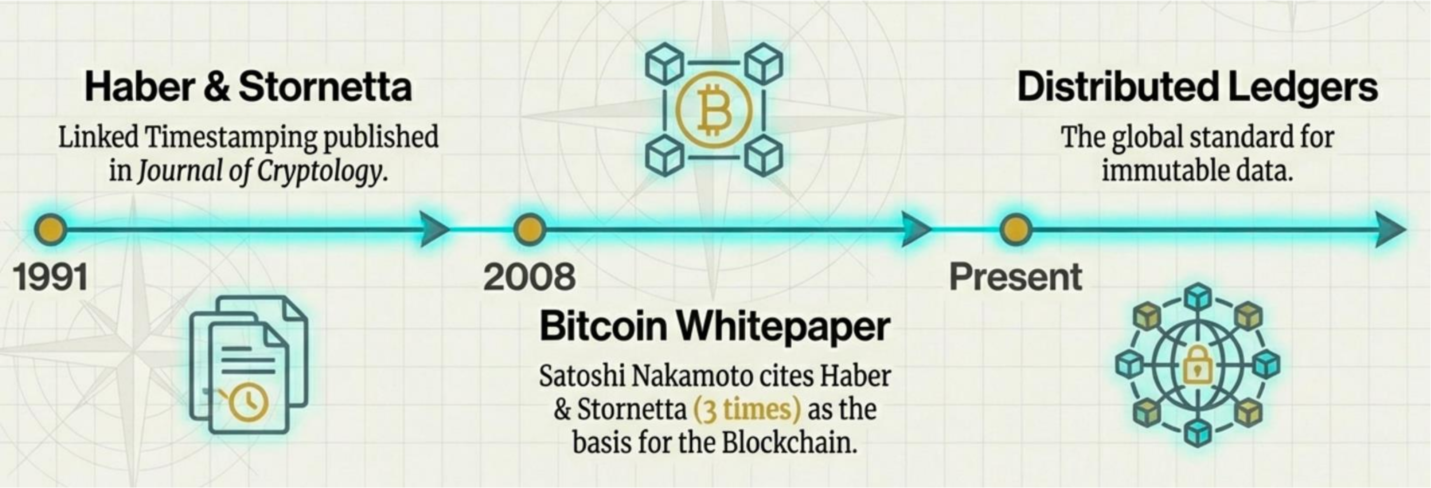
Implications for a Digital Society



The Foundations of Distributed Ledgers

Published in 1991, this work laid the mathematical and structural groundwork for the blockchain revolution. Their solution to the "colluding server" problem is the ancestor of every modern immutable ledger system.



Project Ideas?

Next Class

Oshani will not be attending.

Inwon Kang (Oshani's student) will be presenting the following two papers.

Blockchain Interoperability Landscape	https://ieeexplore.ieee.org/document/10020412
Deciphering Crypto Twitter	https://dl.acm.org/doi/10.1145/3614419.3644026

Please read these papers and ask good questions.

The TA/mentor will note down your class participation for the grade.

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Any Questions?

Please feel free to:

- Message me on the WebEx Space
- Email me at senevo@rpi.edu